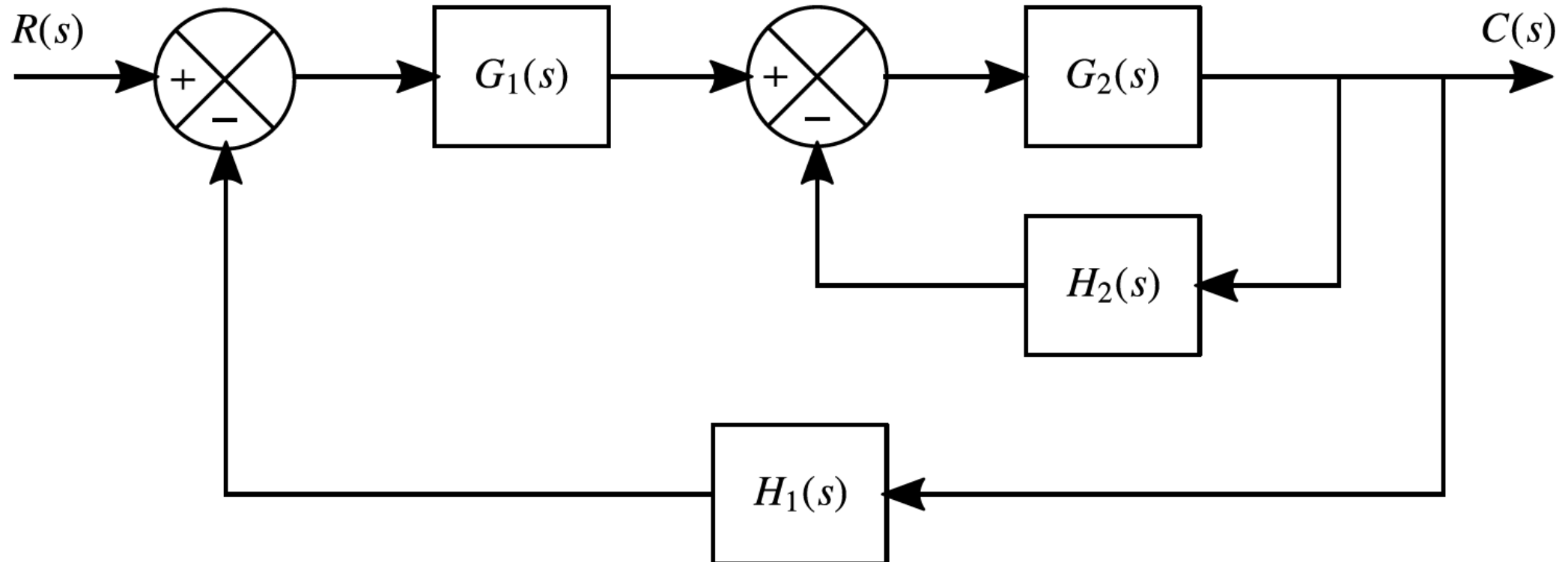
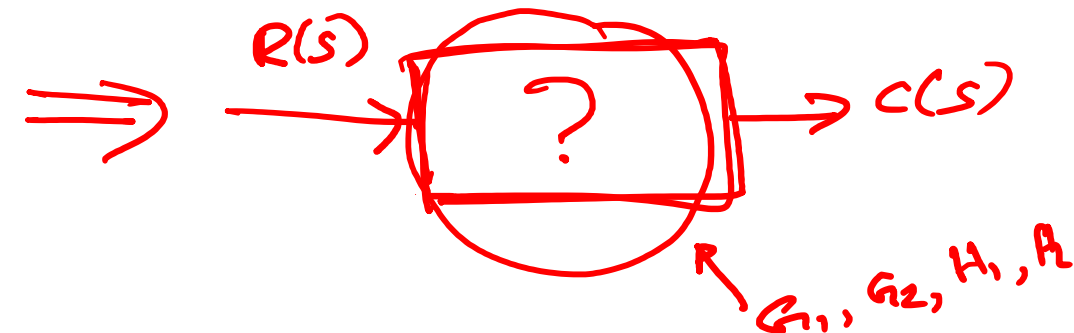
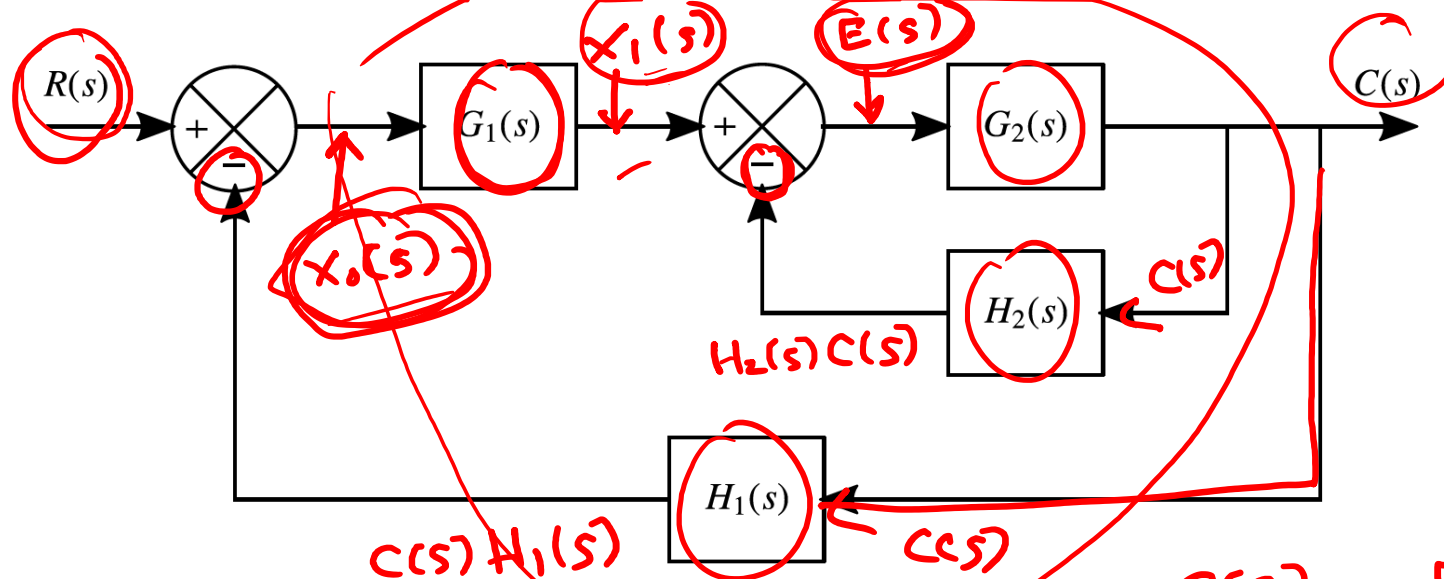


Example: Derive the overall transfer function





Method 1

$$C(s) = E(s) G_2(s)$$

$$C(s) = [x_1(s) - C(s) H_2(s)] G_2(s)$$

$$= [x_0(s) G_1(s) - C(s) H_2(s)] G_2(s)$$

$$C(s) = [(R(s) - C(s) H_1(s)) G_1(s) - C(s) H_2(s)] G_2(s)$$

$$C(s) = E(s) G_2(s)$$

$$E(s) = (x_1(s) - C(s) H_2(s))$$

$$x_1(s) = (x_0(s) G_1(s))$$

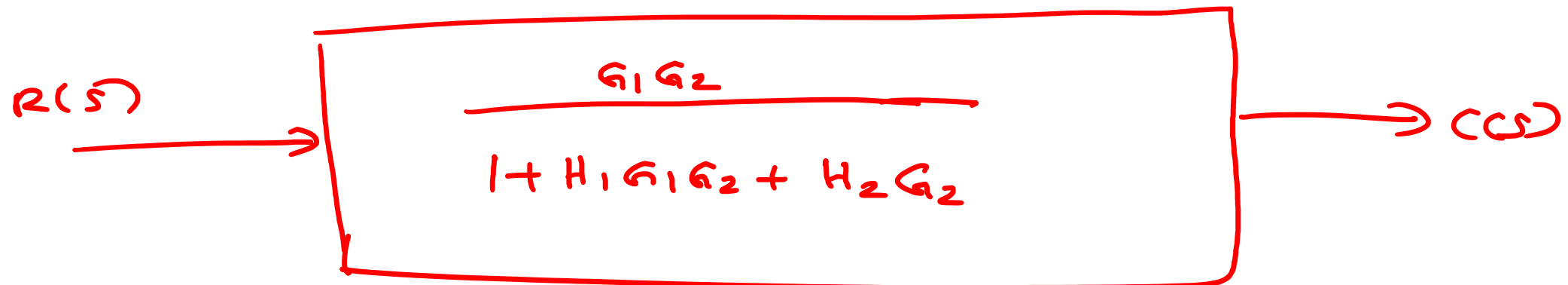
$$x_0(s) = R(s) - C(s) H_1(s)$$

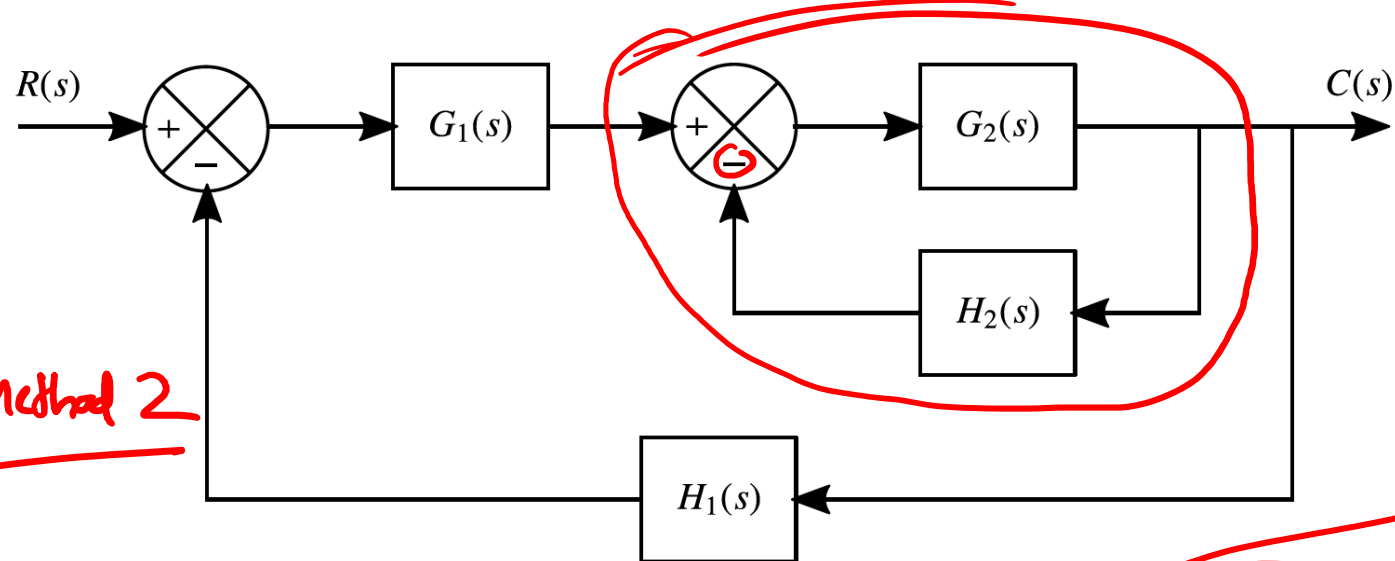
$$\frac{C(s)}{R(s)} = ?$$

$$\underline{C(s)} + \underline{C(s)} H_1(s) G_1(s) G_2(s) + \underline{C(s)} H_2(s) G_2(s) = \underline{R(s)} G_1(s) G_2(s)$$

$$C(s) \left[1 + \cancel{H_1(s) G_1(s)} + H_2(s) \cancel{G_2(s)} \right] = R(s) G_1(s) G_2(s)$$

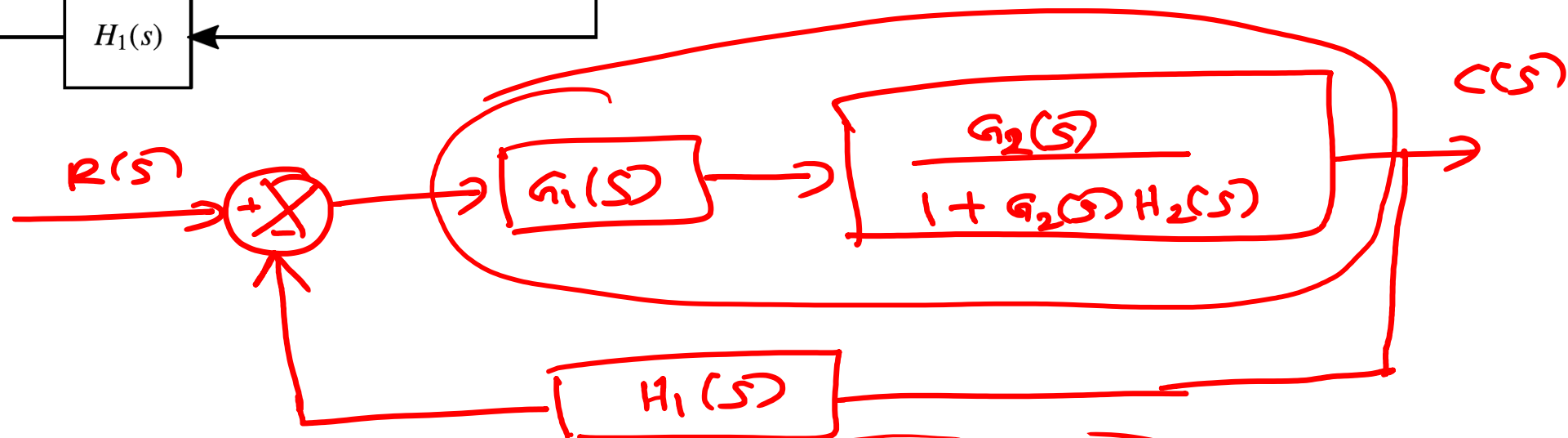
$$\frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s)}{1 + H_1(s) G_1(s) G_2(s) + H_2(s) G_2(s)}$$



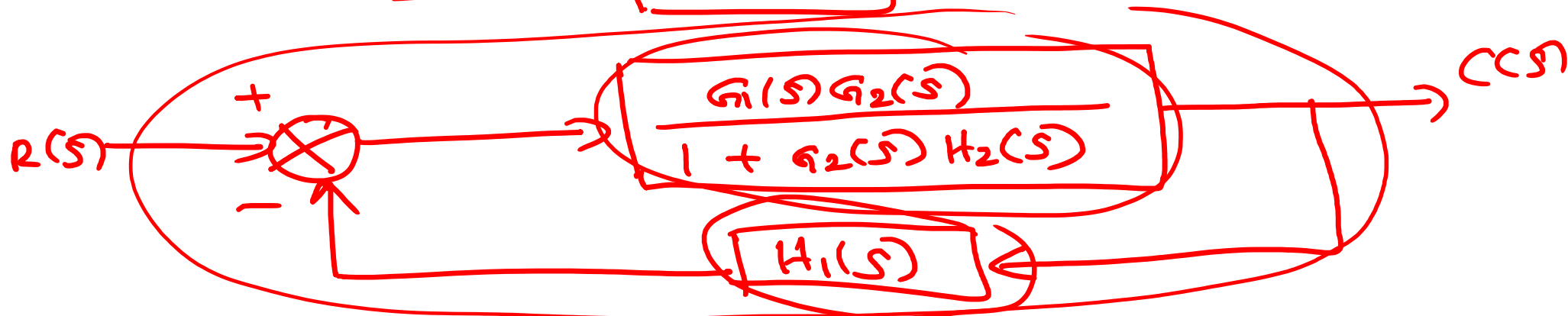


Method 2

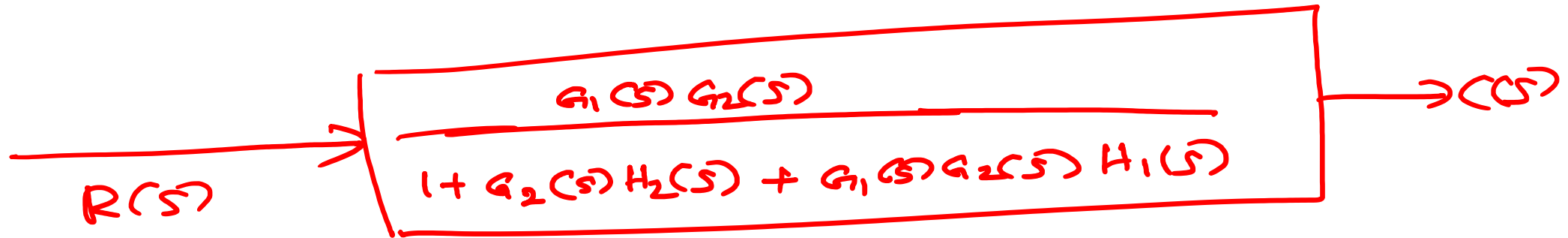
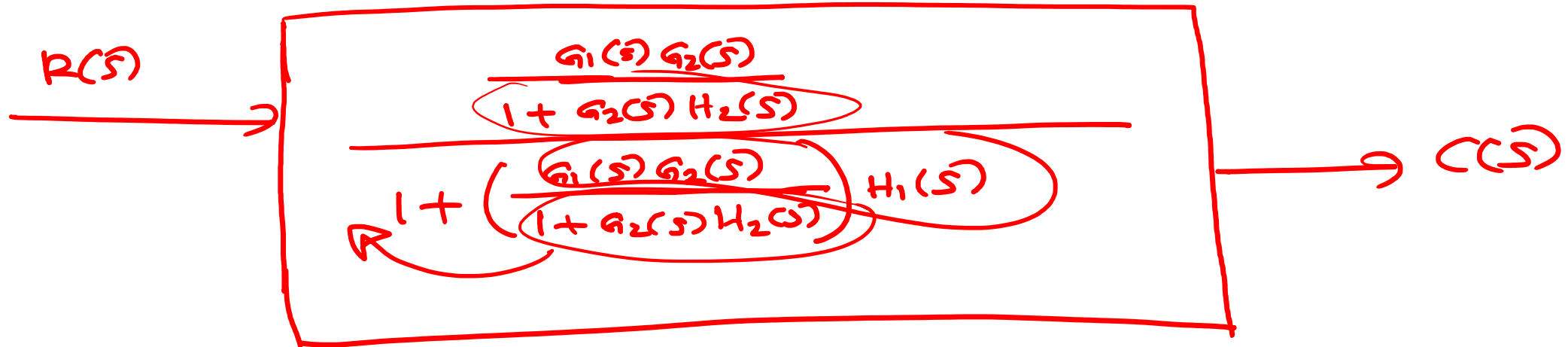
Rule 7



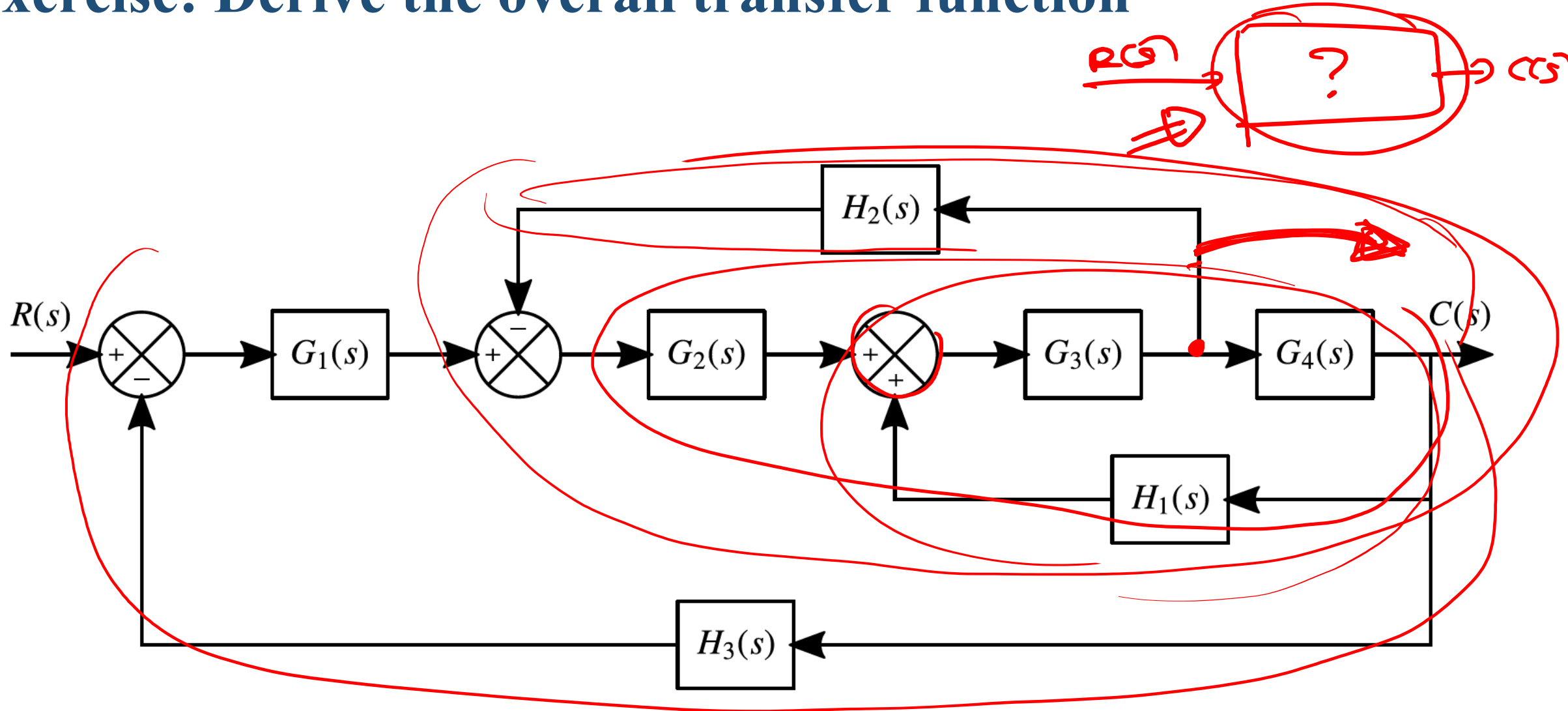
Rule 1

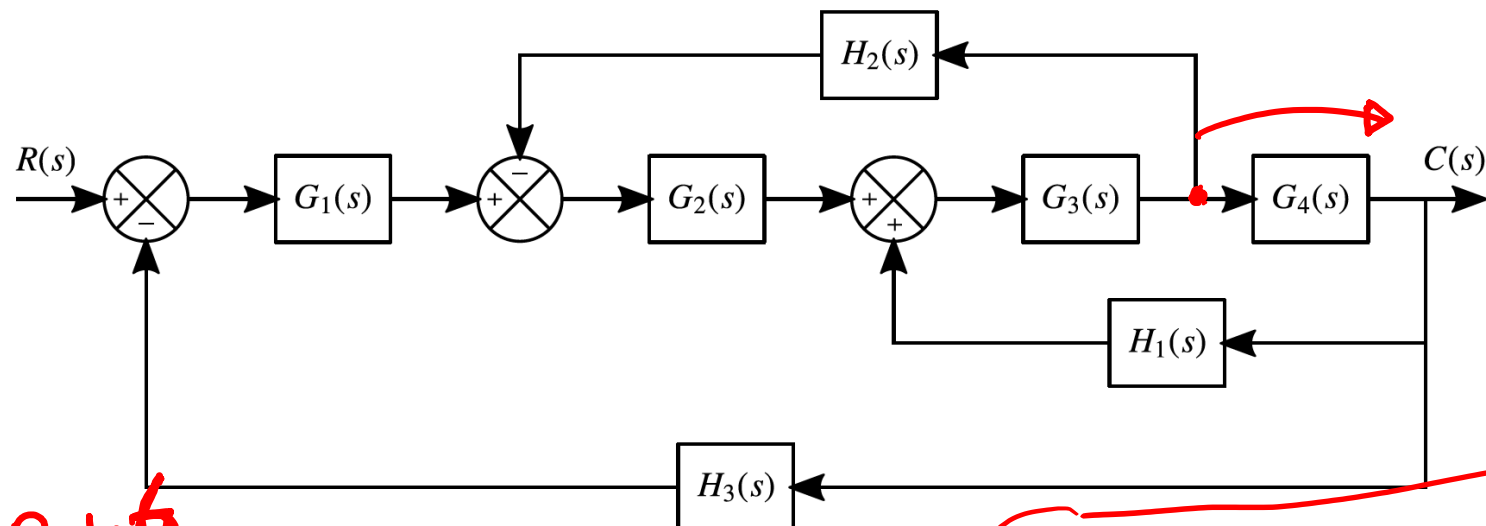


Rule 7

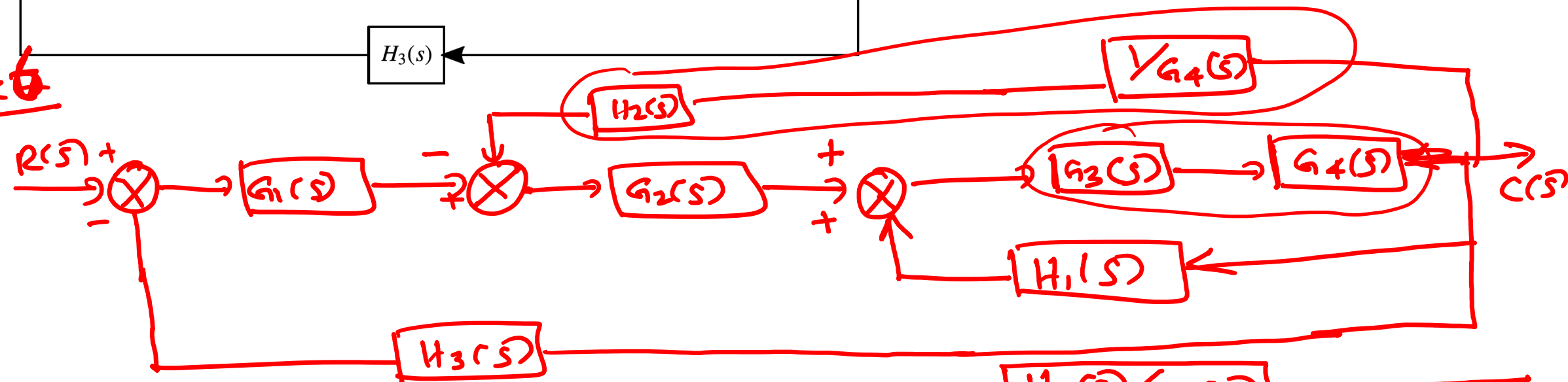


Exercise: Derive the overall transfer function

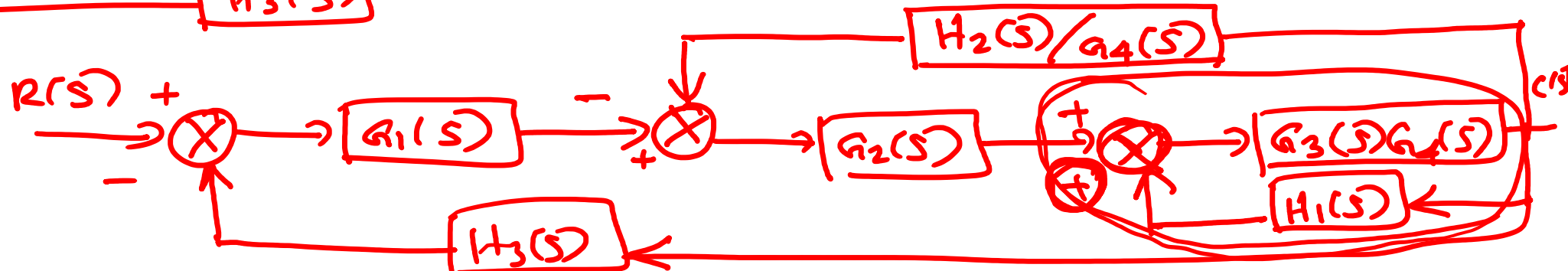




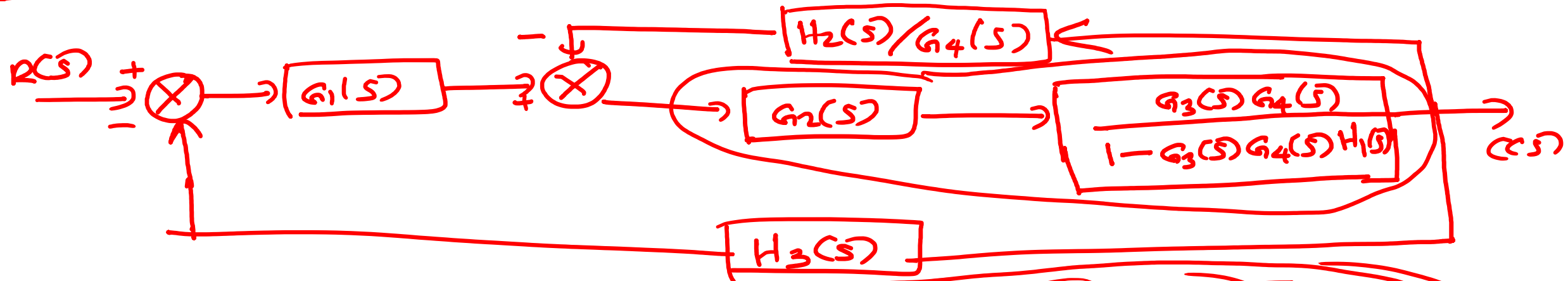
Rule 6



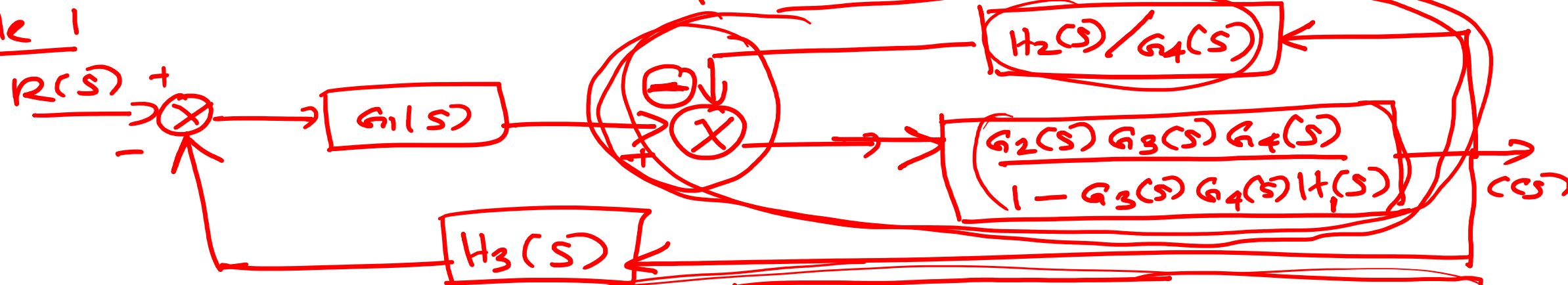
Rule 1



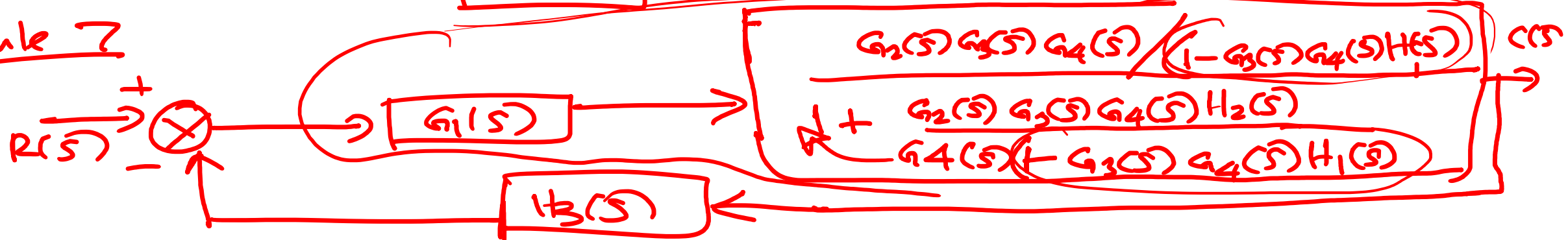
Rule 7



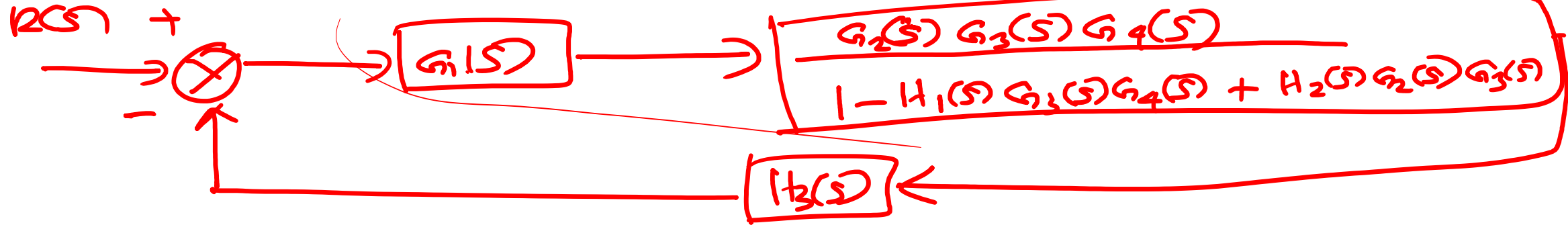
Rule 1



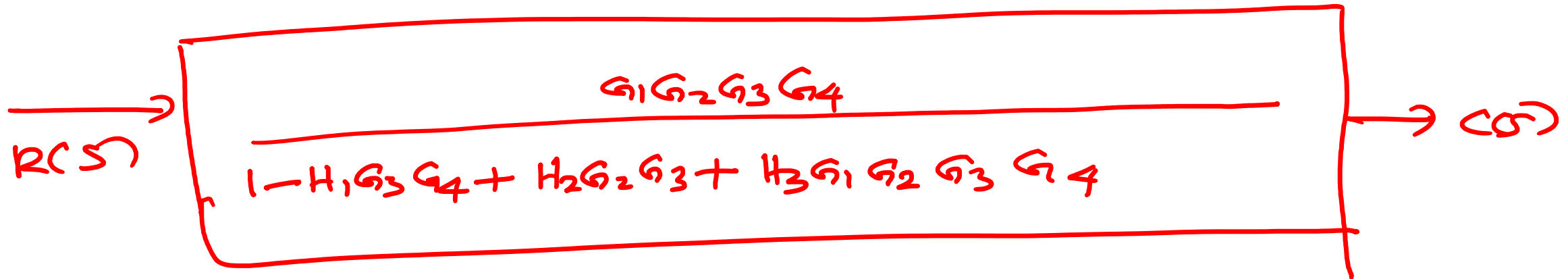
Rule 2



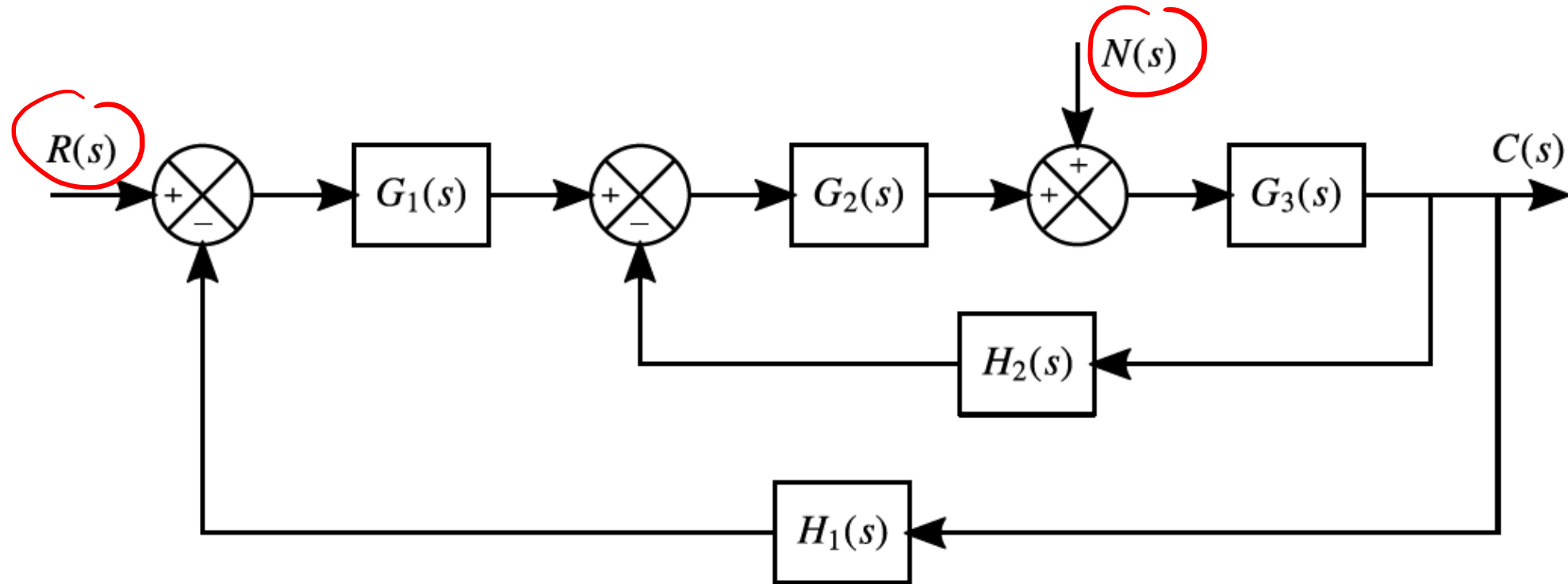
R1

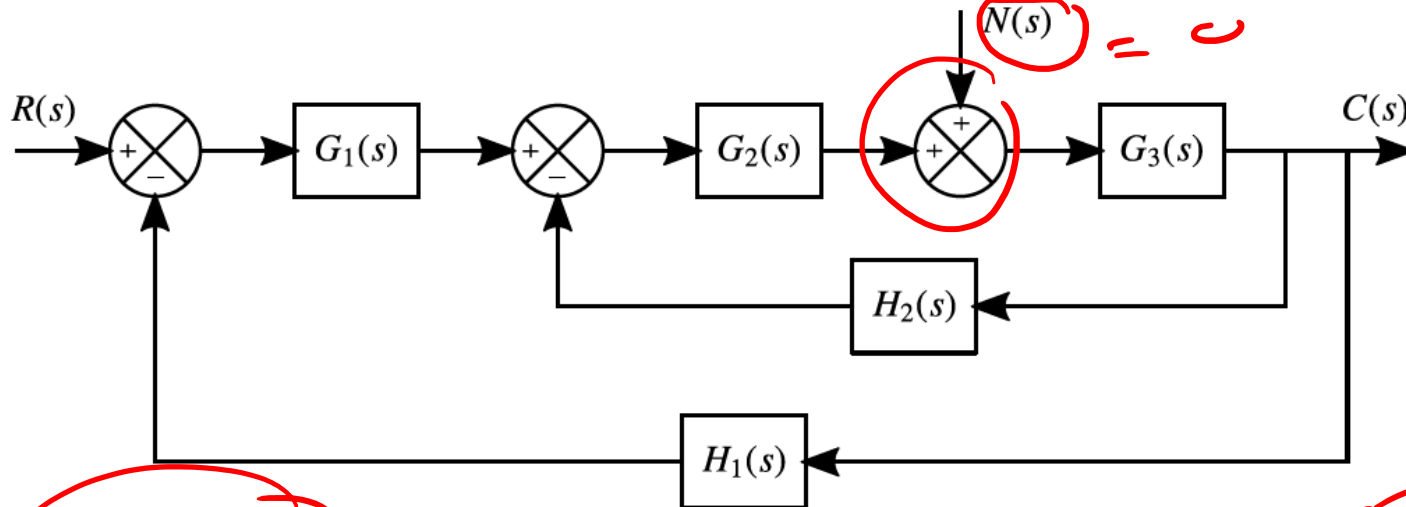


R7

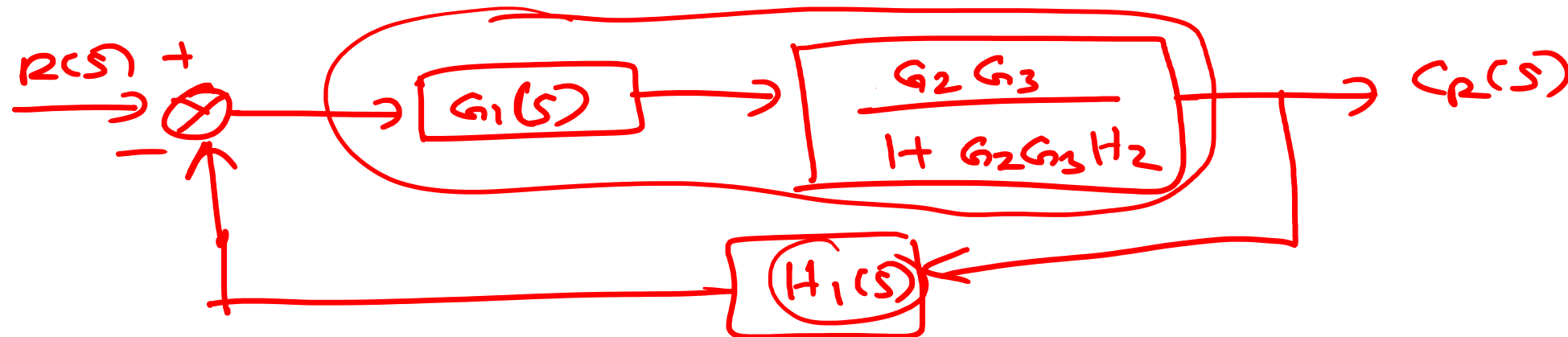
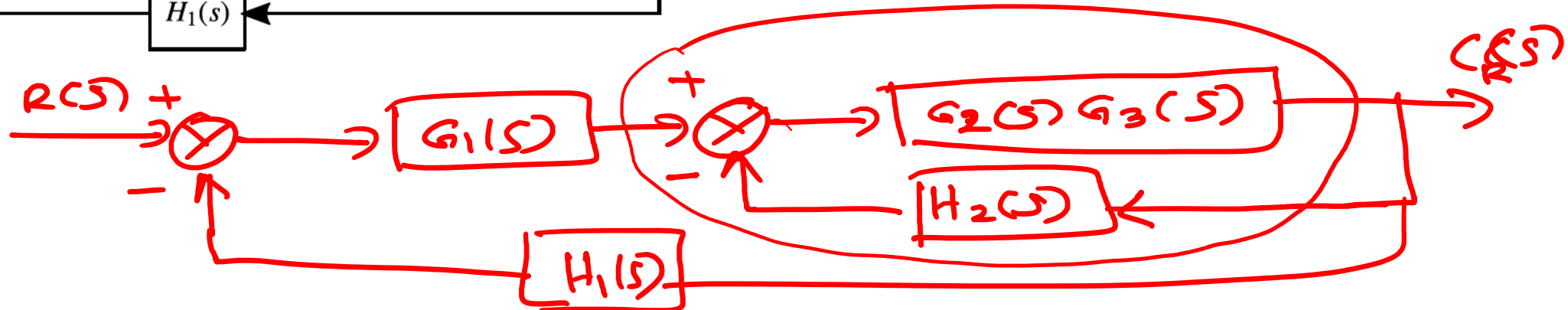


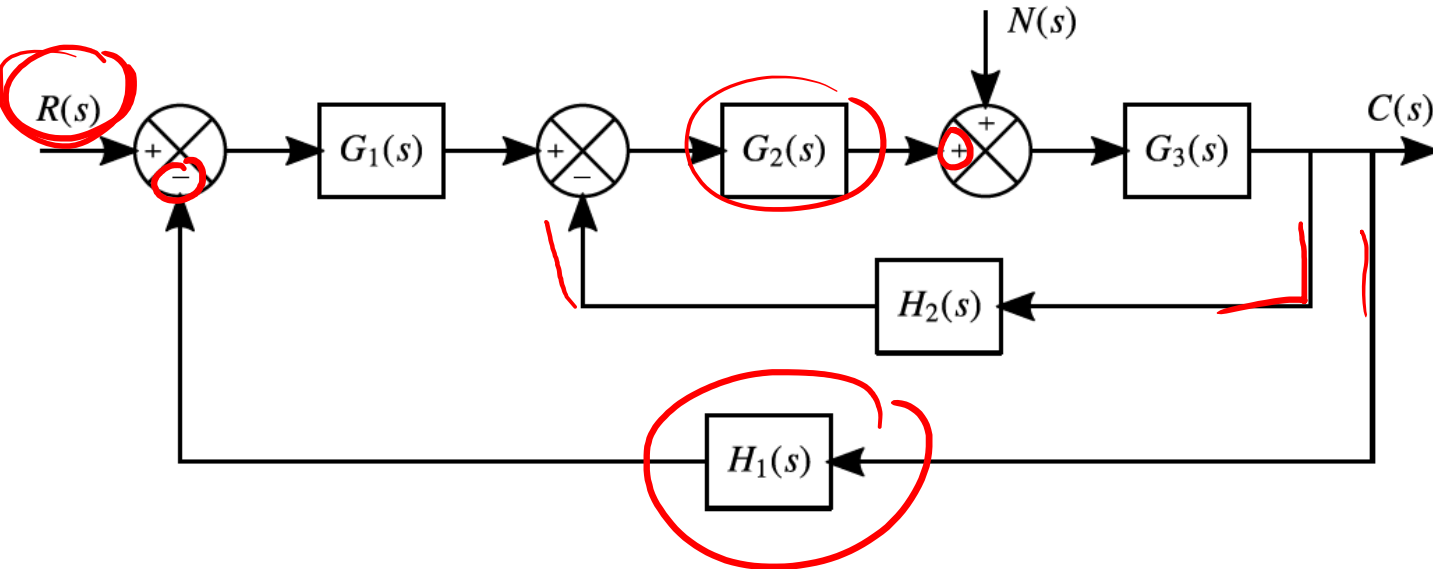
Exercise - Derive the overall transfer function





$N(s) = 0$

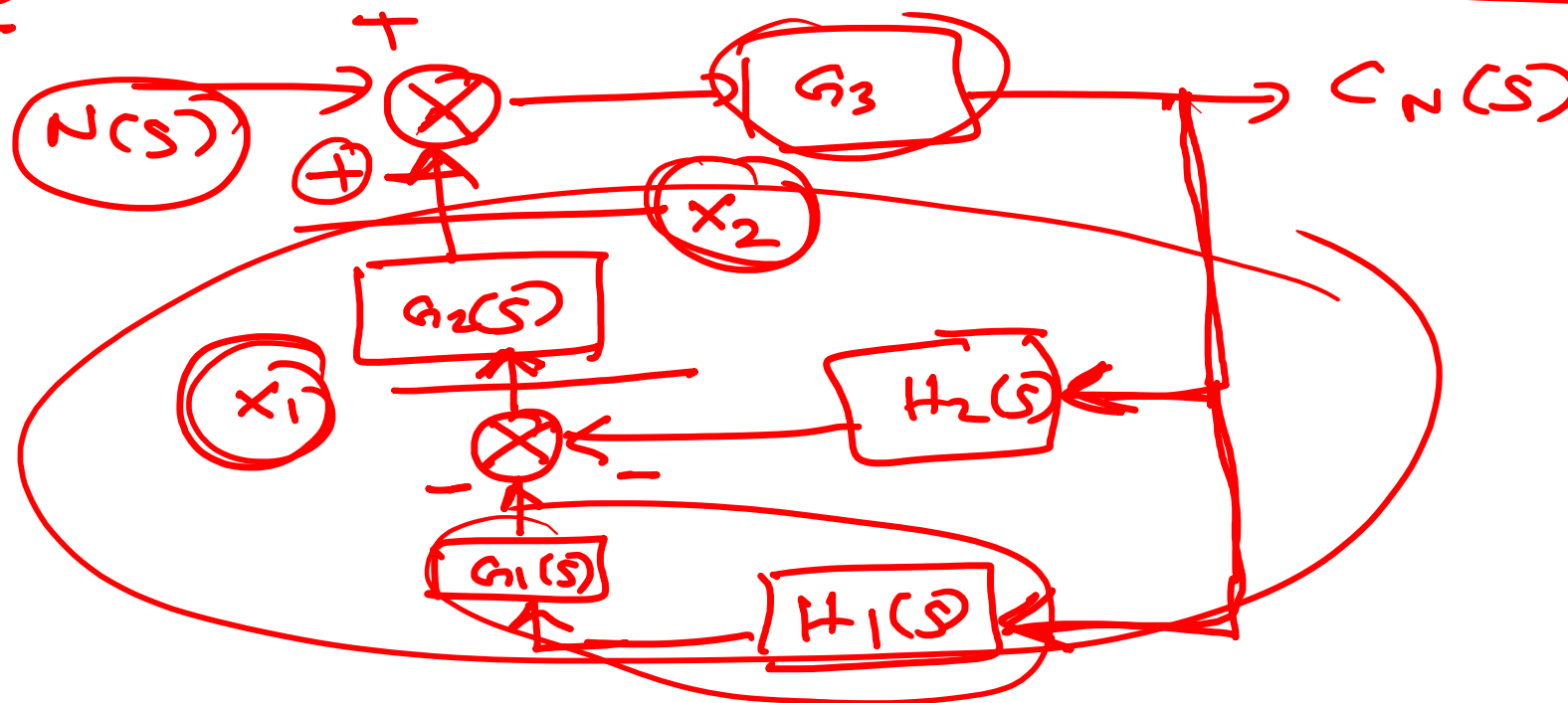




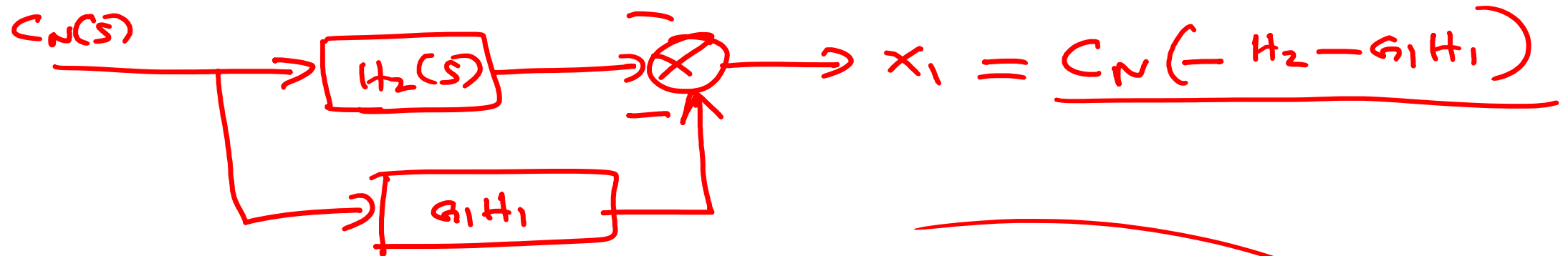
$$\frac{C_R(s)}{R(s)} \neq \frac{G_1 G_2 G_3}{1 + G_1 G_2 G_3 H_1 + H_2 G_2 G_3}$$

$$C_R(s) =$$

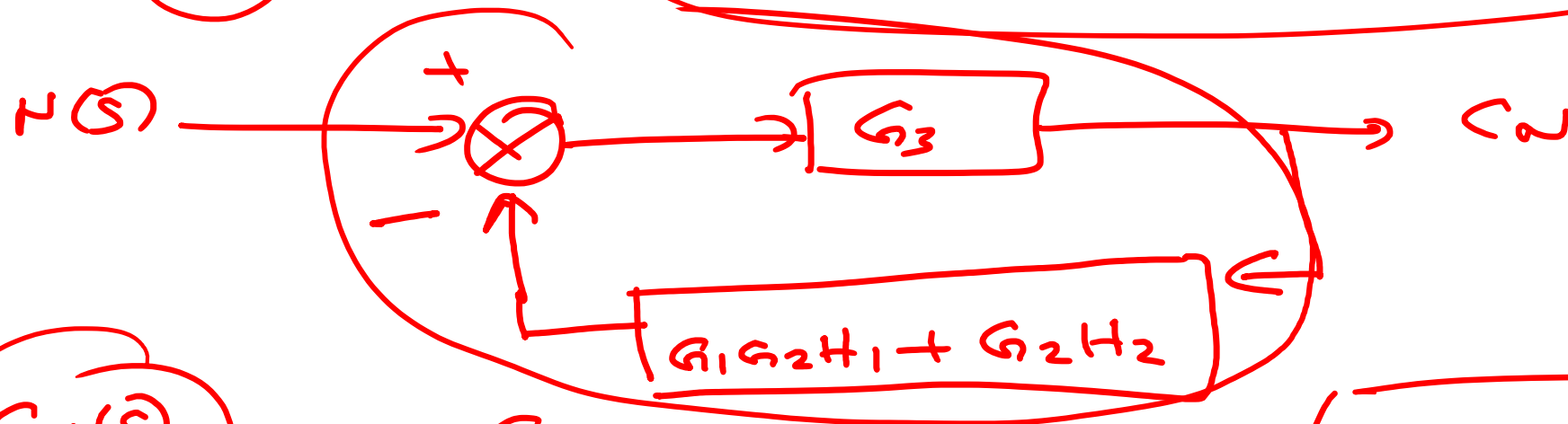
$$\underline{R(s) = 0}$$



$$C(s) = C_p(s) + C_N(s)$$



$$X_2 = X_1 G_2 = (G_1 G_2 H_1 + G_2 H_2) C_N$$



$$\frac{C_N(s)}{N(s)} = \frac{G_3}{1 + G_1 G_2 G_3 H_1 + G_2 G_3 H_2} \Rightarrow C_N(s) = ?$$